

Down Round as Emergent Strategy: Multi-Agent RL Evidence from VC Auctions

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Outline

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What Is a Down Round?

Definition

A *down round* occurs when a company raises capital at a post-money valuation **lower** than its previous round.

- Among 49,996 rounds in our dataset, down rounds constitute **49.35%** of consecutive funding sequences
- Nearly a coin flip — yet the *why* remains poorly understood
- Five distinct mechanisms have been proposed: statistical effects, intertemporal arbitrage, information cascades, market structure, behavioral biases

Core Question: Are down rounds purely statistical noise, or do rational agents *strategically* devalue?

Why Traditional Methods Fall Short

Data Sparsity

Only $\sim 1\%$ of Crunchbase rounds contain valuation data

Identification

Separating statistical, strategic, and behavioral contributions requires counterfactuals unavailable in observational data

Conflicting Theories

Different mechanisms predict similar outcomes — causal attribution is impossible

Computational Randomized Controlled Trial

- Construct a multi-agent RL environment mirroring VC auction economics
- Train agents to learn *optimal* bidding strategies
- Observe what behaviors **emerge** spontaneously

Analogy with LLMs

If down rounds are optimal, they should *emerge* during training — analogous to how chain-of-thought reasoning and in-context learning emerge in large language models brown2020, wei2022.

RL+LLM+Data Framework

Phase A: RL Emergence (this work)

Phase B: DPO-trained LLM Interpreters (planned)

Two-Period Vickrey (Second-Price) Auction

- **Bidders:** $N = 3$ (matching typical VC round investor count)
- **Valuation:** $V \sim \text{LogNormal}(\text{median} = 100, \sigma_{\log} = 1.19)$, calibrated from data
- **Inter-round shift:** $V_1 = V \cdot e^{\mathcal{N}(0, 1.0)}$ — realistic volatility
- **Private signals:** $\theta_i \sim \mathcal{N}(V_t, \sigma^2)$, with $\sigma/V \approx 20\%$
- **Ownership:** Winner acquires $\alpha = 0.17$ (data median)

Key innovation: Stochastic valuation + inter-round shift produce a random baseline DR rate of $\sim 48\%$, matching empirical 49.35% .

MAPPO CTDE Architecture

Decentralized Actor

- 6-dim private observation:
 $[\theta_i, t, P_0, w_i^{prev}, \mu_{belief}, \sigma_{belief}]$
- MLP: $6 \rightarrow 64 \rightarrow 64 \rightarrow 1$
- Gaussian policy: $b_i \sim \mathcal{N}(\mu(\theta_i), \sigma^2)$

Centralized Critic

- 9-dim global state: all signals + ownership
- MLP: $9 \rightarrow 128 \rightarrow 128 \rightarrow 64 \rightarrow 1$
- Observes *everything* during training

Why CTDE?

Models real-world information asymmetry: each VC sees only their private signal, yet training leverages global market information.

Reward Function & Belief Update

Rewards incorporate ownership dilution and cross-round optionality:

$$R_0^{winner} = \alpha \cdot V - P_0 + \gamma \cdot (1 - \alpha) \cdot \alpha \cdot V \cdot \frac{0.5}{N}$$

$$R_1^{winner} = \alpha' \cdot V_1 - P_1, \quad \alpha' = (1 - o_{total}) \cdot \alpha$$

Bayesian belief update after Round 0:

$$\mu_{post} = \frac{\mu_{prior} / \sigma_{prior}^2 + P_0 / \sigma_{obs}^2}{1 / \sigma_{prior}^2 + 1 / \sigma_{obs}^2}$$

Agents treat the observed price as a noisy signal about opponent types.

Calibration: Random Baseline vs. Empirical Data

Table 1: Stochastic V ($\sigma_{\log} = 1.19$) + v_shift ($\sigma_{shift} = 1.0$) produces a baseline closely matching data.

Statistic	Data	Random Baseline
DR rate (consecutive)	49.35%	48%
DR magnitude (mean)	-62.24%	close match
DR magnitude (median)	-68.11%	close match
Ownership median	0.17	0.17

Untrained agents bidding proportionally to signals already approximate the empirical distribution — *without any strategic learning*.

Result 1: Strategic Emergence of Down Rounds

After 10,000 episodes of MAPPO training:

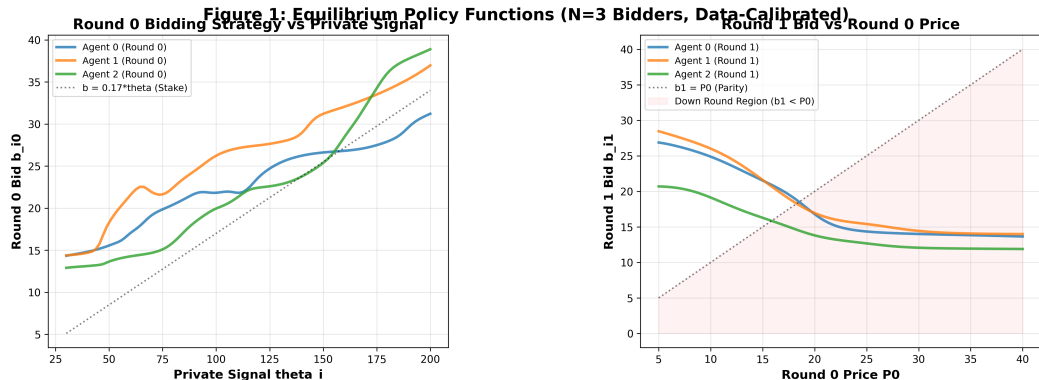
Table 2: DR rate rises from 49% to 82% — a 33% strategic component.

Statistic	Random Baseline	RL-Trained	Strategic Component
DR rate	49%	82%	+33%
DR mag. (mean)	-40%	-29%	milder
DR mag. (median)	-38%	-24%	milder

Interpretation

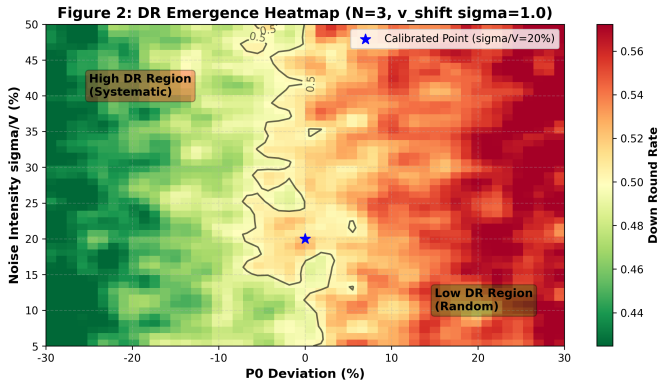
Down rounds arise as *rational optimal strategies* when agents learn to exploit inter-round value shifts and ownership dynamics.

Result 2: Equilibrium Policy Functions



- **Round 0:** bid above stake-value benchmark; **Round 1:** systematic devaluation → down round equilibrium

Result 3: Down Round Emergence Heatmap



- **Phase transition:** high noise $\rightarrow$ systematic DRs across all deviations; calibrated point: DR rate $\approx 76.5\%$

The Down Round Magnitude Paradox

RL-Trained Agents

- DR rate: 82%
- DR magnitude (mean): **-29%**
- DR magnitude (median): **-24%**

Real Market Data

- DR rate: 49.35%
- DR magnitude (mean): **-62%**
- DR magnitude (median): **-68%**

The Paradox

RL agents produce frequent but *mild* DRs; real data shows *severe* DRs.
Strategic rationality alone cannot explain the full severity.

Decomposing Down Rounds

$$DR_{\text{observed}} = \underbrace{DR_{\text{statistical}}}_{\approx 49\%} + \underbrace{DR_{\text{strategic}}}_{\approx 33\%} + \underbrace{DR_{\text{behavioral}}}_{\text{magnitude amplifier}}$$

Behavioral Bias Candidates

- **Loss aversion** kahneman1979: overly conservative bidding after prior losses
- **Anchoring**: over-reliance on P_0 as reference point
- **Winner's curse** thaler1988: fear of overbidding drives defensive underbidding

Behavioral biases likely amplify DR severity *beyond* rational equilibrium levels — this motivates Phase B.

Phase B: DPO-Trained LLM Interpreters

Use DPO rafailov2023 to train LLMs as “decision interpreters”:

- ① **Tokenization:** Convert RL states (signal, price, beliefs) into natural language
- ② **Reasoning chains:** LLM generates “why” explanations for each bid
- ③ **Preference pairs:** Rational reasoning (chosen) vs. biased reasoning (rejected)
- ④ **DPO training:** $\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta \cdot (\log \pi_{\text{rat}} - \log \pi_{\text{bias}}))$

Four Bias Types to Quantify

Anchoring | Loss Aversion ($\lambda = 2.5$) | Overconfidence | Herding

Already generated 1000+ DPO preference samples ($\sim 500\text{KB}$ dataset).

Contributions

Methodological

Computational RCT via RL agents identifies causal mechanisms that observational data cannot separate.

Empirical

Quantification of strategic (+33%) vs. statistical (49%) vs. behavioral (magnitude amplification) down round components.

Conceptual

Bridges RL emergence with LLM post-training interpretability for agentic AI in business research.

- If down rounds are partially *rational equilibrium* outcomes, interventions should target **structural causes**
 - Improve information disclosure to reduce signal noise
 - NOT regulatory restrictions on down rounds themselves
- The strategic component (33%) suggests some DRs are *efficient* market outcomes
- The behavioral component identifies where investor education or decision-support tools could *reduce unnecessary severity*
- Strategic devaluation may be a **feature** of rational optimization, not a bug requiring intervention

Open Problems & Future Work

- ① **Magnitude paradox resolution:** Behavioral bias modeling (Phase B) or alternative reward structures
- ② **Multi-round generalization:** Extend from 2 rounds to realistic 3–7 round sequences
- ③ **LLM-RL alignment:** Can DPO-trained LLMs accurately predict RL actions ($> 80\%$ accuracy)?
- ④ **Endogeneity:** Real-world valuation shifts may be driven by prior-round pricing dynamics (currently exogenous)

Future work: Phase B implementation, multi-round extensions, validation on additional VC datasets

Conclusion

- Down rounds *emerge* as rational optimal strategies in data-calibrated multi-agent RL environments ($N = 3$ bidders)
- Strategic component adds **33%** to the 49% statistical baseline $\rightarrow$ 82% DR rate
- Magnitude paradox: RL agents produce mild DRs (-29%) vs. real data (-62%) $\rightarrow$ behavioral biases amplify severity
- RL+LLM+Data framework bridges emergence with interpretability for agentic AI in business research

Thank you!